

Contents

1	Introduction	1
1.1	Prologue	1
1.2	Automation is everywhere	1
1.3	Contributions	1
1.3.1	Alternative type-class solver in Coq-Elpi	1
1.3.2	Static determinacy checking for Elpi	1
2	State of the Art	3
2.1	The <i>Rocq</i> proof assistant	3
2.1.1	The basis of the language	3
2.1.2	Elaboration	3
2.1.3	Ad-hoc polymorphism: Type Classes	3
2.2	The <i>Elpi</i> language	4
2.2.1	λ -terms representation	4
2.2.2	Hints about unification	4
2.2.3	Controlling backtracking: the ! operator	4
2.2.4	<i>CHR</i> : goal suspension	5
2.3	<i>Coq-Elpi</i> : Rocq and Elpi interleaving	5
2.3.1	Concluding remarks	5
3	Alternative type-classe solver in <i>Coq-Elpi</i>	7
3.1	Why typeclass resolution in <i>Elpi</i> ?	7
3.2	From Rocq goals to <i>Elpi</i> proofs	7
3.2.1	Rule encoding	7
3.2.2	Goal encoding	7
3.3	Alternative search indexing	7
3.3.1	Discrimination trees	7
3.3.2	Some ideas on tabling	7
3.4	Helping hihger-order unification	8
3.4.1	η -equivalent terms	8
3.4.2	β -equivalent terms	8
3.5	Experience report	8
4	Determinacy checking for Elpi	9
4.1	Why determinacy?	9
4.2	Implementation	9
4.2.1	Rule checking	9
4.2.2	Mutual exclusion	9
4.2.3	Caveats	9
4.2.4	Experience report	9

4.3	Formalization in Rocq	9
4.3.1	The language	9
4.3.2	The tree semantics	10
4.3.3	The stack semantics	10
4.3.4	Semantics equivalence	10
4.3.5	The main theorem	10
5	Conclusion and Perspectives	11
	Bibliography	13

CHAPTER 1

Introduction

1.1 Prologue

Automation everywhere in CS, we want to set up boilerplates to help our development.

Examples:

- Continuous integration
- Macros are everywhere
- Artificial intelligence

We delegate tasks to some tools.

expert system →

example plus frappant:

artificial intelligence: based on probability but not reproducible, not predicatable

several other examples in computer sciences:

...

1.2 Automation is everywhere

In the domain of formal proof automation is everywhere, and it is formally called elaboration [51]

In formal proof, proofs are verbose and have lot of details: types, coercions, canonical structures, type-classes.

In theorem provers the user benefit of notations, implicit inference and tactics to ease his job.

Simply typed lambda calculus is an example of automation: an algo checks if ...

1.3 Contributions

In this manuscript we focus on an alternative type-class solver written in the elpi meta-language.

We also talk about a formalized tool...

1.3.1 Alternative type-class solver in Coq-Elpi

1.3.2 Static determinacy checking for Elpi

CHAPTER 2

State of the Art

At the very base of my PhD there are two softwares, the Rocq Proof assistant (formerly known as *Coq* [6]) and the *Elpi* [13].

2.1 The *Rocq* proof assistant

[6]

Theorem provers with HO logic programming languages: [15]

Mathematical components: [32]

Abella: [21]

SSReflect: [22]

Isabelle: [38]

2.1.1 The basis of the language

2.1.2 Elaboration

In fig. 2.1 (taken from [51]), we have a skemtical representation of the elaboration process. [51]

Coercions: [4]

2.1.3 Ad-hoc polymorphism: Type Classes

One of the most interesting fragment of elaboration is [53, 47, 54]

Cite Haskell [25], in caml it does not exists: `+. vs +`

IRIS: [29]

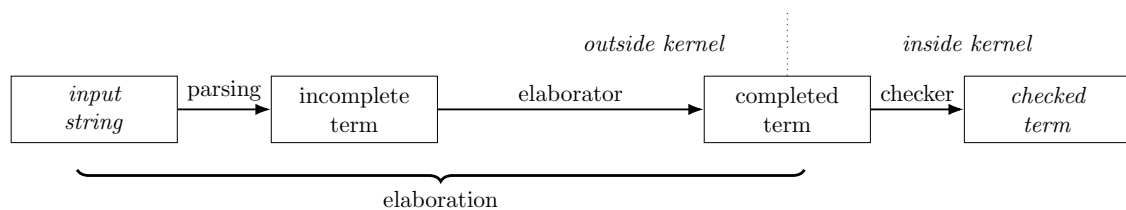


Figure 2.1: Elaboration skeme

```

Class Eqb (T : Type) := {eqb : T -> T -> bool}.

Instance eqNat : Eqb nat := {eqb a b := Nat.eqb a b}.
Instance eqProd (A: Type) (B: Type) : Eqb A -> Eqb B -> Eqb (A * B) :=
  {eqb a b := eqb a.1 b.1 && eqb a.2 b.2}.

```

(a) Eqb typeclass in Rocq

```

class Eqb a where
  eqb :: a -> a -> Bool

instance Eqb Integer where eqb = integerEq

instance (Eqb a, Eqb b) => Eqb (a, b) where
  eqb (x1, y1) (x2, y2) = eqb x1 x2 && eqb y1 y2

```

(b) Eqb typeclass in Haskell

Figure 2.2: Eqb typeclass in different languages

2.2 The *Elpi* language

Elpi [13] is a dialect of λ prolog [34, 37]

Teyjus: [36, 35]

Elf: [39]

Twelf: [41]

Lambda prolog: [44]

Mercury: [46]

WAM: [3]

Prolog semantics: [11]

2.2.1 λ -terms representation

HOAS: [40]

2.2.2 Hints about unification

Miller: [34]

Huet: [28]

Modes: [55]

2.2.3 Controlling backtracking: the ! operator

SWI prolog: [48]

Andrews: [2]

Vink: [10]

2.2.4 *CHR*: goal suspension

[[20](#), [19](#)]

CHR elpi: [[24](#), section 4.3]

CHR Pfenning: [[33](#)]

2.3 *Coq-Elpi*: Rocq and Elpi interleaving

Coqelpi: [[49](#)]

Hierarchy Builder: [[9](#)]

Trackt: [[7](#)]

Equality test [[23](#)]

Derive: [[50](#)]

2.3.1 Concluding remarks

Overall, Elpi is good for elaboration

Alternative type-classes solver in *Coq-Elpi*

Typeclasses are an elegant way to implement ad-hoc polymorphism [52], so that, the same symbol can be reused in different context while keeping the same mathematical properties. From an implementation point of view, typeclasses are

3.1 Why typeclass resolution in *Elpi*?

[16]

Rule base automation!

3.2 From Rocq goals to *Elpi* proofs

3.2.1 Rule encoding

The hooks

3.2.2 Goal encoding

3.3 Alternative search indexing

3.3.1 Discrimination trees

Discrimination trees: Pfenning notes *

TODO: do some benchmarks

Discrimination trees: opt for lists and tree depths.

3.3.2 Some ideas on tabling

[45, 42]

*<https://www.cs.cmu.edu/~fp/courses/99-atp/lectures/lecture28.html>

3.4 Helping higher-order unification

[17]

?: [14]

Variadic functions are typechecked in coq [8]

3.4.1 η -equivalent terms

3.4.2 β -equivalent terms

[1]

Beta problema outside the pattern can come back in LL [36]

3.5 Experience report

IRIS: [29]

CHAPTER 4

Determinacy checking for Elpi

4.1 Why determinacy?

[[12](#), [46](#), [26](#), [27](#), [31](#), [30](#)]

4.2 Implementation

4.2.1 Rule checking

4.2.2 Mutual exclusion

Using discrimination tree

4.2.3 Caveats

For example, mutual exclusion is not perfect: over approximation of unification

Rule checking ignores flexible rule loading

4.2.4 Experience report

4.3 Formalization in Rocq

Using ssreflect [[22](#)]

4.3.1 The language

Input must come before outputs

Inputs must come before outputs, with Disc tree this is not mandatory, since unification is weaker.

EG:

```
f 3 3 (g X X).
```

```
f 3 4 (g 3 4).
```

```
main:-
```

```
  f X Y (g X Y).
```

In the code above, both rules can be used with the query, but, the discrimination tree, save the world

4.3.2 The tree semantics

4.3.3 The stack semantics

[\[43\]](#)

4.3.4 Semantics equivalence

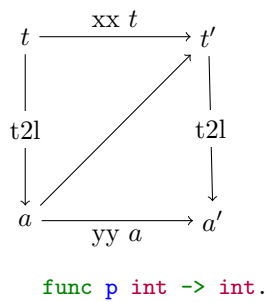
[\[5\]](#)

4.3.5 The main theorem

CHAPTER 5

Conclusion and Perspectives

[18]



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